Square H4: CQKUD collected expectations with numerical values

First six states at $\epsilon = 0.5,\ 0.8,\ 1.2$

Only the collected final expression is shown for each state. The number beneath each expectation or paired bracket is its value in Hartree; full-precision coefficients and complex moments are in the accompanying JSON. Here $H = \Lambda\bar{H}$ with $\Lambda =$ `5.7176103816932775` Hartree, $V = e^{-i\epsilon\bar{H}}$, and $X_\epsilon = \sin(\epsilon\bar{H}) = (V^\dagger - V)/(2i)$. The solver uses $U_\epsilon = iV$ internally; this display converts to the unshifted notation of the reference PDF. The common denominator $4^n \prod_{j=1}^{n} \beta_j^2$ is divided into every displayed coefficient; the JSON also retains the original numerator coefficients and denominator. Printed values are rounded; final values use unrounded `Decimal` spectral sums for the saved float64 Hamiltonian.

$\epsilon = 0.5$

Iteration $n = 0$, $\epsilon = 0.5$
4. Collected final expression

$$\langle q_0|H|q_0\rangle \simeq \underbrace{\langle\Phi_0|H|\Phi_0\rangle}_{\approx -1.7139986422219549103}$$

$$\simeq -1.71399864222195 \text{ Hartree.}$$

Iteration $n = 1$, $\epsilon = 0.5$
4. Collected final expression

$$\langle q_1|H|q_1\rangle \simeq 569.0327900720766\ \underbrace{\langle\Phi_0|H|\Phi_0\rangle}_{\approx -1.7139986422219549103} - 162.6272732871267\,i\ \underbrace{\left(\langle\Phi_0|H(V^\dagger)^1|\Phi_0\rangle - \langle\Phi_0|HV^1|\Phi_0\rangle\right)}_{\approx 0.53284571232561824927\,i}$$

$$- 272.3790682842995\ \underbrace{\left(\langle\Phi_0|H(V^\dagger)^2|\Phi_0\rangle + \langle\Phi_0|HV^2|\Phi_0\rangle\right)}_{\approx -3.2585459375096481863}$$

$$\simeq -1.10647786459087 \text{ Hartree.}$$

Iteration $n = 2$, $\epsilon = 0.5$
4. Collected final expression

$$\langle q_2|H|q_2\rangle \simeq 279395.0084809545\ \underbrace{\langle\Phi_0|H|\Phi_0\rangle}_{\approx -1.7139986422219549103} - 125151.5238270513\,i\ \underbrace{\left(\langle\Phi_0|H(V^\dagger)^1|\Phi_0\rangle - \langle\Phi_0|HV^1|\Phi_0\rangle\right)}_{\approx 0.53284571232561824927\,i}$$

$$- 181326.5639720599\ \underbrace{\left(\langle\Phi_0|H(V^\dagger)^2|\Phi_0\rangle + \langle\Phi_0|HV^2|\Phi_0\rangle\right)}_{\approx -3.2585459375096481863} + 40980.42948697482\,i\ \underbrace{\left(\langle\Phi_0|H(V^\dagger)^3|\Phi_0\rangle - \langle\Phi_0|HV^3|\Phi_0\rangle\right)}_{\approx 1.5438524551682598373\,i}$$

$$+ 41689.69464403519\ \underbrace{\left(\langle\Phi_0|H(V^\dagger)^4|\Phi_0\rangle + \langle\Phi_0|HV^4|\Phi_0\rangle\right)}_{\approx -2.7680199835889209941}$$

$$\simeq -0.91848567184919 \text{ Hartree.}$$

Iteration $n = 3$, $\epsilon = 0.5$
4. Collected final expression

$$\begin{aligned}
\langle q_3|H|q_3\rangle \simeq\ & 298571281.4337836 \underbrace{\langle\Phi_0|H|\Phi_0\rangle}_{\approx -1.7139986422219549103} - 168034178.7626228\,i \underbrace{\left(\langle\Phi_0|H(V^\dagger)^1|\Phi_0\rangle - \langle\Phi_0|HV^1|\Phi_0\rangle\right)}_{\approx 0.53284571232561824927\,i} \\
& - 219583376.1397285 \underbrace{\left(\langle\Phi_0|H(V^\dagger)^2|\Phi_0\rangle + \langle\Phi_0|HV^2|\Phi_0\rangle\right)}_{\approx -3.2585459375096481863} + 82905599.94200149\,i \underbrace{\left(\langle\Phi_0|H(V^\dagger)^3|\Phi_0\rangle - \langle\Phi_0|HV^3|\Phi_0\rangle\right)}_{\approx 1.5438524551682598373\,i} \\
& + 82680769.83493157 \underbrace{\left(\langle\Phi_0|H(V^\dagger)^4|\Phi_0\rangle + \langle\Phi_0|HV^4|\Phi_0\rangle\right)}_{\approx -2.7680199835889209941} - 16140706.18129165\,i \underbrace{\left(\langle\Phi_0|H(V^\dagger)^5|\Phi_0\rangle - \langle\Phi_0|HV^5|\Phi_0\rangle\right)}_{\approx 2.3966598674003788273\,i} \\
& - 12382757.11647446 \underbrace{\left(\langle\Phi_0|H(V^\dagger)^6|\Phi_0\rangle + \langle\Phi_0|HV^6|\Phi_0\rangle\right)}_{\approx -2.0079707503756156320} \\
\simeq\ & -1.38556292512746 \text{ Hartree.}
\end{aligned}$$

Iteration $n = 4$, $\epsilon = 0.5$
4. Collected final expression

$$\begin{aligned}
\langle q_4|H|q_4\rangle \simeq\ & 291675182959.2208 \underbrace{\langle\Phi_0|H|\Phi_0\rangle}_{\approx -1.7139986422219549103} - 203232875228.1910\,i \underbrace{\left(\langle\Phi_0|H(V^\dagger)^1|\Phi_0\rangle - \langle\Phi_0|HV^1|\Phi_0\rangle\right)}_{\approx 0.53284571232561824927\,i} \\
& - 228950674869.1096 \underbrace{\left(\langle\Phi_0|H(V^\dagger)^2|\Phi_0\rangle + \langle\Phi_0|HV^2|\Phi_0\rangle\right)}_{\approx -3.2585459375096481863} + 120185519588.2699\,i \underbrace{\left(\langle\Phi_0|H(V^\dagger)^3|\Phi_0\rangle - \langle\Phi_0|HV^3|\Phi_0\rangle\right)}_{\approx 1.5438524551682598373\,i} \\
& + 107979763911.6118 \underbrace{\left(\langle\Phi_0|H(V^\dagger)^4|\Phi_0\rangle + \langle\Phi_0|HV^4|\Phi_0\rangle\right)}_{\approx -2.7680199835889209941} - 38905855968.12261\,i \underbrace{\left(\langle\Phi_0|H(V^\dagger)^5|\Phi_0\rangle - \langle\Phi_0|HV^5|\Phi_0\rangle\right)}_{\approx 2.3966598674003788273\,i} \\
& - 27841759358.24048 \underbrace{\left(\langle\Phi_0|H(V^\dagger)^6|\Phi_0\rangle + \langle\Phi_0|HV^6|\Phi_0\rangle\right)}_{\approx -2.0079707503756156320} + 5315041882.192464\,i \underbrace{\left(\langle\Phi_0|H(V^\dagger)^7|\Phi_0\rangle - \langle\Phi_0|HV^7|\Phi_0\rangle\right)}_{\approx 3.0046767553685994014\,i} \\
& + 2975081898.205348 \underbrace{\left(\langle\Phi_0|H(V^\dagger)^8|\Phi_0\rangle + \langle\Phi_0|HV^8|\Phi_0\rangle\right)}_{\approx -1.0580870097656434729} \\
\simeq\ & -1.45706219215062 \text{ Hartree.}
\end{aligned}$$

Iteration $n = 5$, $\epsilon = 0.5$
4. Collected final expression

$$\langle q_5|H|q_5\rangle \simeq 1261917677922614 \underbrace{\langle\Phi_0|H|\Phi_0\rangle}_{\approx -1.7139986422219549103} - 1004015886949390\, i \underbrace{\left(\langle\Phi_0|H(V^\dagger)^1|\Phi_0\rangle - \langle\Phi_0|HV^1|\Phi_0\rangle\right)}_{\approx 0.53284571232561824927\, i}$$

$$- 1032911405093890 \underbrace{\left(\langle\Phi_0|H(V^\dagger)^2|\Phi_0\rangle + \langle\Phi_0|HV^2|\Phi_0\rangle\right)}_{\approx -3.2585459375096481863} + 659265474807230.8\, i \underbrace{\left(\langle\Phi_0|H(V^\dagger)^3|\Phi_0\rangle - \langle\Phi_0|HV^3|\Phi_0\rangle\right)}_{\approx 1.5438524551682598373\, i}$$

$$+ 558662442345972.4 \underbrace{\left(\langle\Phi_0|H(V^\dagger)^4|\Phi_0\rangle + \langle\Phi_0|HV^4|\Phi_0\rangle\right)}_{\approx -2.7680199835889209941} - 274015380419518.9\, i \underbrace{\left(\langle\Phi_0|H(V^\dagger)^5|\Phi_0\rangle - \langle\Phi_0|HV^5|\Phi_0\rangle\right)}_{\approx 2.3966598674003788273\, i}$$

$$- 190335504470127.1 \underbrace{\left(\langle\Phi_0|H(V^\dagger)^6|\Phi_0\rangle + \langle\Phi_0|HV^6|\Phi_0\rangle\right)}_{\approx -2.0079707503756156320} + 65375277021886.67\, i \underbrace{\left(\langle\Phi_0|H(V^\dagger)^7|\Phi_0\rangle - \langle\Phi_0|HV^7|\Phi_0\rangle\right)}_{\approx 3.0046767553685994014\, i}$$

$$+ 36595559474084.17 \underbrace{\left(\langle\Phi_0|H(V^\dagger)^8|\Phi_0\rangle + \langle\Phi_0|HV^8|\Phi_0\rangle\right)}_{\approx -1.0580870097656434729} - 6814510486177.674\, i \underbrace{\left(\langle\Phi_0|H(V^\dagger)^9|\Phi_0\rangle - \langle\Phi_0|HV^9|\Phi_0\rangle\right)}_{\approx 3.3074530328231895645\, i}$$

$$- 2969931049143.802 \underbrace{\left(\langle\Phi_0|H(V^\dagger)^{10}|\Phi_0\rangle + \langle\Phi_0|HV^{10}|\Phi_0\rangle\right)}_{\approx -0.017571627431013454598}$$

$$\simeq -0.55793093694726 \text{ Hartree.}$$

$\epsilon = 0.8$

Iteration $n = 0$, $\epsilon = 0.8$
4. Collected final expression

$$\langle q_0|H|q_0\rangle \simeq \underbrace{\langle \Phi_0|H|\Phi_0\rangle}_{\approx -1.7139986422219549103}$$

$$\simeq -1.71399864222195 \text{ Hartree.}$$

Iteration $n = 1$, $\epsilon = 0.8$
4. Collected final expression

$$\langle q_1|H|q_1\rangle \simeq 242.9114136046073 \underbrace{\langle \Phi_0|H|\Phi_0\rangle}_{\approx -1.7139986422219549103} - 103.6083708740815\, i \underbrace{\left(\langle \Phi_0|H(V^\dagger)^1|\Phi_0\rangle - \langle \Phi_0|HV^1|\Phi_0\rangle\right)}_{\approx 0.84681618762395919122\, i}$$

$$- 109.1637426189016 \underbrace{\left(\langle \Phi_0|H(V^\dagger)^2|\Phi_0\rangle + \langle \Phi_0|HV^2|\Phi_0\rangle\right)}_{\approx -3.0001662989749354033}$$

$$\simeq -1.10320579379105 \text{ Hartree.}$$

Iteration $n = 2$, $\epsilon = 0.8$
4. Collected final expression

$$\langle q_2|H|q_2\rangle \simeq 51401.68277308659 \underbrace{\langle \Phi_0|H|\Phi_0\rangle}_{\approx -1.7139986422219549103} - 32351.41813668650\, i \underbrace{\left(\langle \Phi_0|H(V^\dagger)^1|\Phi_0\rangle - \langle \Phi_0|HV^1|\Phi_0\rangle\right)}_{\approx 0.84681618762395919122\, i}$$

$$- 32243.42107665950 \underbrace{\left(\langle \Phi_0|H(V^\dagger)^2|\Phi_0\rangle + \langle \Phi_0|HV^2|\Phi_0\rangle\right)}_{\approx -3.0001662989749354033} + 10315.27159295141\, i \underbrace{\left(\langle \Phi_0|H(V^\dagger)^3|\Phi_0\rangle - \langle \Phi_0|HV^3|\Phi_0\rangle\right)}_{\approx 2.3211444136241064810\, i}$$

$$+ 6603.890080019638 \underbrace{\left(\langle \Phi_0|H(V^\dagger)^4|\Phi_0\rangle + \langle \Phi_0|HV^4|\Phi_0\rangle\right)}_{\approx -1.8302240980000421244}$$

$$\simeq -0.91843039460536 \text{ Hartree.}$$

Iteration $n = 3$, $\epsilon = 0.8$
4. Collected final expression

$$\begin{aligned}\langle q_3|H|q_3\rangle \simeq\ & 24208253.63367886 \underbrace{\langle\Phi_0|H|\Phi_0\rangle}_{\approx -1.7139986422219549103} - 18115311.49769285\,i \underbrace{\left(\langle\Phi_0|H(V^\dagger)^1|\Phi_0\rangle - \langle\Phi_0|HV^1|\Phi_0\rangle\right)}_{\approx 0.84681618762395919122\,i} \\ & - 17415733.73988726 \underbrace{\left(\langle\Phi_0|H(V^\dagger)^2|\Phi_0\rangle + \langle\Phi_0|HV^2|\Phi_0\rangle\right)}_{\approx -3.0001662989749354033} + 8767718.842726072\,i \underbrace{\left(\langle\Phi_0|H(V^\dagger)^3|\Phi_0\rangle - \langle\Phi_0|HV^3|\Phi_0\rangle\right)}_{\approx 2.3211444136241064810\,i} \\ & + 6103233.231930318 \underbrace{\left(\langle\Phi_0|H(V^\dagger)^4|\Phi_0\rangle + \langle\Phi_0|HV^4|\Phi_0\rangle\right)}_{\approx -1.8302240980000421244} - 1640292.827004470\,i \underbrace{\left(\langle\Phi_0|H(V^\dagger)^5|\Phi_0\rangle - \langle\Phi_0|HV^5|\Phi_0\rangle\right)}_{\approx 3.1967799598152527156\,i} \\ & - 791338.7634277226 \underbrace{\left(\langle\Phi_0|H(V^\dagger)^6|\Phi_0\rangle + \langle\Phi_0|HV^6|\Phi_0\rangle\right)}_{\approx -0.22777467475063368023} \\ \simeq\ & -1.38202314486757 \text{ Hartree.}\end{aligned}$$

Iteration $n = 4$, $\epsilon = 0.8$
4. Collected final expression

$$\begin{aligned}\langle q_4|H|q_4\rangle \simeq\ & 11027501250.41170 \underbrace{\langle\Phi_0|H|\Phi_0\rangle}_{\approx -1.7139986422219549103} - 9405203021.465950\,i \underbrace{\left(\langle\Phi_0|H(V^\dagger)^1|\Phi_0\rangle - \langle\Phi_0|HV^1|\Phi_0\rangle\right)}_{\approx 0.84681618762395919122\,i} \\ & - 8501494660.752115 \underbrace{\left(\langle\Phi_0|H(V^\dagger)^2|\Phi_0\rangle + \langle\Phi_0|HV^2|\Phi_0\rangle\right)}_{\approx -3.0001662989749354033} + 5453464509.415681\,i \underbrace{\left(\langle\Phi_0|H(V^\dagger)^3|\Phi_0\rangle - \langle\Phi_0|HV^3|\Phi_0\rangle\right)}_{\approx 2.3211444136241064810\,i} \\ & + 3785168292.508527 \underbrace{\left(\langle\Phi_0|H(V^\dagger)^4|\Phi_0\rangle + \langle\Phi_0|HV^4|\Phi_0\rangle\right)}_{\approx -1.8302240980000421244} - 1694956086.875232\,i \underbrace{\left(\langle\Phi_0|H(V^\dagger)^5|\Phi_0\rangle - \langle\Phi_0|HV^5|\Phi_0\rangle\right)}_{\approx 3.1967799598152527156\,i} \\ & - 873857473.0292991 \underbrace{\left(\langle\Phi_0|H(V^\dagger)^6|\Phi_0\rangle + \langle\Phi_0|HV^6|\Phi_0\rangle\right)}_{\approx -0.22777467475063368023} + 217055578.1958137\,i \underbrace{\left(\langle\Phi_0|H(V^\dagger)^7|\Phi_0\rangle - \langle\Phi_0|HV^7|\Phi_0\rangle\right)}_{\approx 3.2556510859480033290\,i} \\ & + 76436450.57471710 \underbrace{\left(\langle\Phi_0|H(V^\dagger)^8|\Phi_0\rangle + \langle\Phi_0|HV^8|\Phi_0\rangle\right)}_{\approx 1.3860417407686303962} \\ \simeq\ & -1.46212365372230 \text{ Hartree.}\end{aligned}$$

Iteration $n = 5$, $\epsilon = 0.8$
4. Collected final expression

$$\begin{aligned}
\langle q_5|H|q_5\rangle \simeq\ & 21038298705621.55 \underbrace{\langle\Phi_0|H|\Phi_0\rangle}_{\approx -1.7139986422219549103} - 19141811491861.93\,i \underbrace{\left(\langle\Phi_0|H(V^\dagger)^1|\Phi_0\rangle - \langle\Phi_0|HV^1|\Phi_0\rangle\right)}_{\approx 0.84681618762395919122\,i} \\
& - 16969666858828.47 \underbrace{\left(\langle\Phi_0|H(V^\dagger)^2|\Phi_0\rangle + \langle\Phi_0|HV^2|\Phi_0\rangle\right)}_{\approx -3.0001662989749354033} + 12328198570183.05\,i \underbrace{\left(\langle\Phi_0|H(V^\dagger)^3|\Phi_0\rangle - \langle\Phi_0|HV^3|\Phi_0\rangle\right)}_{\approx 2.3211444136241064810\,i} \\
& + 8764513504128.838 \underbrace{\left(\langle\Phi_0|H(V^\dagger)^4|\Phi_0\rangle + \langle\Phi_0|HV^4|\Phi_0\rangle\right)}_{\approx -1.8302240980000421244} - 4923893531160.532\,i \underbrace{\left(\langle\Phi_0|H(V^\dagger)^5|\Phi_0\rangle - \langle\Phi_0|HV^5|\Phi_0\rangle\right)}_{\approx 3.1967799598152527156\,i} \\
& - 2742033648054.741 \underbrace{\left(\langle\Phi_0|H(V^\dagger)^6|\Phi_0\rangle + \langle\Phi_0|HV^6|\Phi_0\rangle\right)}_{\approx -0.22777467475063368023} + 1103125714896.390\,i \underbrace{\left(\langle\Phi_0|H(V^\dagger)^7|\Phi_0\rangle - \langle\Phi_0|HV^7|\Phi_0\rangle\right)}_{\approx 3.2556510859480033290\,i} \\
& + 456803378096.7257 \underbrace{\left(\langle\Phi_0|H(V^\dagger)^8|\Phi_0\rangle + \langle\Phi_0|HV^8|\Phi_0\rangle\right)}_{\approx 1.3860417407686303962} - 105023249797.8273\,i \underbrace{\left(\langle\Phi_0|H(V^\dagger)^9|\Phi_0\rangle - \langle\Phi_0|HV^9|\Phi_0\rangle\right)}_{\approx 2.4986378782400982213\,i} \\
& - 28765556682.44267 \underbrace{\left(\langle\Phi_0|H(V^\dagger)^{10}|\Phi_0\rangle + \langle\Phi_0|HV^{10}|\Phi_0\rangle\right)}_{\approx 2.5930713674867377749} \\
\simeq\ & -0.55852558678781 \text{ Hartree.}
\end{aligned}$$

$\epsilon = 1.2$

Iteration $n = 0$, $\epsilon = 1.2$
4. Collected final expression

$$\langle q_0|H|q_0\rangle \simeq \underbrace{\langle \Phi_0|H|\Phi_0\rangle}_{\approx -1.7139986422219549103}$$

$$\simeq -1.71399864222195 \text{ Hartree.}$$

Iteration $n = 1$, $\epsilon = 1.2$
4. Collected final expression

$$\langle q_1|H|q_1\rangle \simeq 127.5771595399004 \underbrace{\langle \Phi_0|H|\Phi_0\rangle}_{\approx -1.7139986422219549103} - 71.87701023081387\, i \underbrace{\left(\langle \Phi_0|H(V^\dagger)^1|\Phi_0\rangle - \langle \Phi_0|HV^1|\Phi_0\rangle\right)}_{\approx 1.2526854953542934527\, i}$$

$$- 51.16752656960491 \underbrace{\left(\langle \Phi_0|H(V^\dagger)^2|\Phi_0\rangle + \langle \Phi_0|HV^2|\Phi_0\rangle\right)}_{\approx -2.4924303724391603781}$$

$$\simeq -1.09629275967935 \text{ Hartree.}$$

Iteration $n = 2$, $\epsilon = 1.2$
4. Collected final expression

$$\langle q_2|H|q_2\rangle \simeq 14025.68291105770 \underbrace{\langle \Phi_0|H|\Phi_0\rangle}_{\approx -1.7139986422219549103} - 10739.53162642800\, i \underbrace{\left(\langle \Phi_0|H(V^\dagger)^1|\Phi_0\rangle - \langle \Phi_0|HV^1|\Phi_0\rangle\right)}_{\approx 1.2526854953542934527\, i}$$

$$- 8359.666528914322 \underbrace{\left(\langle \Phi_0|H(V^\dagger)^2|\Phi_0\rangle + \langle \Phi_0|HV^2|\Phi_0\rangle\right)}_{\approx -2.4924303724391603781} + 3256.026540634638\, i \underbrace{\left(\langle \Phi_0|H(V^\dagger)^3|\Phi_0\rangle - \langle \Phi_0|HV^3|\Phi_0\rangle\right)}_{\approx 3.0494213477204835595\, i}$$

$$+ 1409.561663314479 \underbrace{\left(\langle \Phi_0|H(V^\dagger)^4|\Phi_0\rangle + \langle \Phi_0|HV^4|\Phi_0\rangle\right)}_{\approx -0.22777467475063368023}$$

$$\simeq -0.91850149256893 \text{ Hartree.}$$

Iteration $n = 3$, $\epsilon = 1.2$
4. Collected final expression

$$\begin{aligned}\langle q_3|H|q_3\rangle \simeq\ & 3537903.193418975 \underbrace{\langle\Phi_0|H|\Phi_0\rangle}_{\approx -1.7139986422219549103} - 3034503.327170646\, i \underbrace{\left(\langle\Phi_0|H(V^\dagger)^1|\Phi_0\rangle - \langle\Phi_0|HV^1|\Phi_0\rangle\right)}_{\approx 1.2526854953542934527\, i} \\ & - 2471570.954845848 \underbrace{\left(\langle\Phi_0|H(V^\dagger)^2|\Phi_0\rangle + \langle\Phi_0|HV^2|\Phi_0\rangle\right)}_{\approx -2.4924303724391603781} + 1418868.027600545\, i \underbrace{\left(\langle\Phi_0|H(V^\dagger)^3|\Phi_0\rangle - \langle\Phi_0|HV^3|\Phi_0\rangle\right)}_{\approx 3.0494213477204835595\, i} \\ & + 783182.4526791019 \underbrace{\left(\langle\Phi_0|H(V^\dagger)^4|\Phi_0\rangle + \langle\Phi_0|HV^4|\Phi_0\rangle\right)}_{\approx -0.22777467475063368023} - 246399.1622506350\, i \underbrace{\left(\langle\Phi_0|H(V^\dagger)^5|\Phi_0\rangle - \langle\Phi_0|HV^5|\Phi_0\rangle\right)}_{\approx 3.1378145020559871415\, i} \\ & - 80252.86625789131 \underbrace{\left(\langle\Phi_0|H(V^\dagger)^6|\Phi_0\rangle + \langle\Phi_0|HV^6|\Phi_0\rangle\right)}_{\approx 2.0631812248568705017} \\ \simeq\ & -1.37447786833031 \text{ Hartree.}\end{aligned}$$

Iteration $n = 4$, $\epsilon = 1.2$
4. Collected final expression

$$\begin{aligned}\langle q_4|H|q_4\rangle \simeq\ & 907921739.6257937 \underbrace{\langle\Phi_0|H|\Phi_0\rangle}_{\approx -1.7139986422219549103} - 829982727.6292947\, i \underbrace{\left(\langle\Phi_0|H(V^\dagger)^1|\Phi_0\rangle - \langle\Phi_0|HV^1|\Phi_0\rangle\right)}_{\approx 1.2526854953542934527\, i} \\ & - 684956043.7892475 \underbrace{\left(\langle\Phi_0|H(V^\dagger)^2|\Phi_0\rangle + \langle\Phi_0|HV^2|\Phi_0\rangle\right)}_{\approx -2.4924303724391603781} + 465816875.5323918\, i \underbrace{\left(\langle\Phi_0|H(V^\dagger)^3|\Phi_0\rangle - \langle\Phi_0|HV^3|\Phi_0\rangle\right)}_{\approx 3.0494213477204835595\, i} \\ & + 284195202.6474495 \underbrace{\left(\langle\Phi_0|H(V^\dagger)^4|\Phi_0\rangle + \langle\Phi_0|HV^4|\Phi_0\rangle\right)}_{\approx -0.22777467475063368023} - 135078346.0989742\, i \underbrace{\left(\langle\Phi_0|H(V^\dagger)^5|\Phi_0\rangle - \langle\Phi_0|HV^5|\Phi_0\rangle\right)}_{\approx 3.1378145020559871415\, i} \\ & - 56860074.93344345 \underbrace{\left(\langle\Phi_0|H(V^\dagger)^6|\Phi_0\rangle + \langle\Phi_0|HV^6|\Phi_0\rangle\right)}_{\approx 2.0631812248568705017} + 15396038.79520822\, i \underbrace{\left(\langle\Phi_0|H(V^\dagger)^7|\Phi_0\rangle - \langle\Phi_0|HV^7|\Phi_0\rangle\right)}_{\approx 1.5194797906825721355\, i} \\ & + 3663678.548436862 \underbrace{\left(\langle\Phi_0|H(V^\dagger)^8|\Phi_0\rangle + \langle\Phi_0|HV^8|\Phi_0\rangle\right)}_{\approx 3.0909550315191541096} \\ \simeq\ & -1.47241138632798 \text{ Hartree.}\end{aligned}$$

Iteration $n = 5$, $\epsilon = 1.2$
4. Collected final expression

$$\langle q_5|H|q_5\rangle \simeq 851540924248.8696 \underbrace{\langle\Phi_0|H|\Phi_0\rangle}_{\approx -1.7139986422219549103} - 799357137809.9429\, i \underbrace{\left(\langle\Phi_0|H(V^\dagger)^1|\Phi_0\rangle - \langle\Phi_0|HV^1|\Phi_0\rangle\right)}_{\approx 1.2526854953542934527\, i}$$

$$- 675199236271.8854 \underbrace{\left(\langle\Phi_0|H(V^\dagger)^2|\Phi_0\rangle + \langle\Phi_0|HV^2|\Phi_0\rangle\right)}_{\approx -2.4924303724391603781} + 499643907514.9662\, i \underbrace{\left(\langle\Phi_0|H(V^\dagger)^3|\Phi_0\rangle - \langle\Phi_0|HV^3|\Phi_0\rangle\right)}_{\approx 3.0494213477204835595\, i}$$

$$+ 330378237745.7863 \underbrace{\left(\langle\Phi_0|H(V^\dagger)^4|\Phi_0\rangle + \langle\Phi_0|HV^4|\Phi_0\rangle\right)}_{\approx -0.22777467475063368023} - 187400734101.0384\, i \underbrace{\left(\langle\Phi_0|H(V^\dagger)^5|\Phi_0\rangle - \langle\Phi_0|HV^5|\Phi_0\rangle\right)}_{\approx 3.1378145020559871415\, i}$$

$$- 93413506031.73143 \underbrace{\left(\langle\Phi_0|H(V^\dagger)^6|\Phi_0\rangle + \langle\Phi_0|HV^6|\Phi_0\rangle\right)}_{\approx 2.0631812248568705017} + 37872520928.04617\, i \underbrace{\left(\langle\Phi_0|H(V^\dagger)^7|\Phi_0\rangle - \langle\Phi_0|HV^7|\Phi_0\rangle\right)}_{\approx 1.5194797906825721355\, i}$$

$$+ 13032399229.08118 \underbrace{\left(\langle\Phi_0|H(V^\dagger)^8|\Phi_0\rangle + \langle\Phi_0|HV^8|\Phi_0\rangle\right)}_{\approx 3.0909550315191541096} - 3076378741.747490\, i \underbrace{\left(\langle\Phi_0|H(V^\dagger)^9|\Phi_0\rangle - \langle\Phi_0|HV^9|\Phi_0\rangle\right)}_{\approx -0.81981676818577920035\, i}$$

$$- 568178368.2332921 \underbrace{\left(\langle\Phi_0|H(V^\dagger)^{10}|\Phi_0\rangle + \langle\Phi_0|HV^{10}|\Phi_0\rangle\right)}_{\approx 2.3360743621070288388}$$

$$\simeq -0.56013051367801 \text{ Hartree.}$$